Linear equations

1

a

i $y = 2x - 7$ ii Yes

b

i $y = \frac{x}{2} + 8$ ii Yes

c

i $y = \frac{-1}{x} + 6$ ii No

d

i $y + \frac{x}{16} = 14$ ii Yes

Tables of values

2

a

i The y-value increases by 6.

ii The y-value is equal to six times the x-value.

iii $y = 6x$

b

i The y-value increases by 1.

ii The y-value is equal to the x-value plus seven.

iii $y = x + 7$

c

i The y-value decreases by 1.

ii The y-value is equal to the x-value multiplied by -1 and then minus six.

iii $y = -x - 6$

d

i The y-value decreases by 4.

ii The y-value is equal to negative four times the x-value plus two.

iii $y = -4x + 2$

e

i The y-value increases by 2.

ii The y-value is equal to two times the x-value plus five.

iii $y = 2x + 5$

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3

a Yes

b -7 units

c Because there is a consistent change in y for every one unit change in x.

d $y = -7x - 5$

e The change in y-value is the number that goes in front of the x in the rule.

4

a Yes **b** Yes **c** No **d** Yes

e Yes **f** No **g** Yes **h** Yes

i No **j** Yes **k** Yes **l** No

m Yes **n** Yes **o** No **p** No

q No **r** Yes

5

a

i 0 **ii** $y = 4x$

iii

x	1	2	3	4	5	14
y	4	8	12	16	20	56

b

i -3 **ii** $y = 2x - 3$

iii

x	1	2	3	4	5	30
y	-1	1	3	5	7	57

c

i 0 **ii** $y = -2x$

iii

x	-16	1	2	3	4	5
y	32	-2	-4	-6	-8	-10

d

i 60 **ii** $y = -8x + 60$

iii

x	-30	1	2	3	4	5
y	300	52	44	36	28	20

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e

i -6 ii $y = -7x - 6$

iii

x	1	2	3	4	5	30
y	-13	-20	-27	-34	-41	-216

6

a

i $m = 2, c = 0$ ii $y = 2x$

iii

x	0	1	2	3	4	5	21
y	0	2	4	6	8	10	42

b

i $m = 5, c = 8$ ii $y = 5x + 8$

iii

x	0	1	2	3	4	5	29
y	8	13	18	23	28	33	153

c

i $m = 5, c = -21$ ii $y = 5x - 21$

iii

x	0	1	2	3	4	5	21
y	-21	-16	-11	-6	-1	4	84

d

i $m = -3, c = 24$ ii $y = -3x + 24$

iii

x	0	1	2	3	4	5	65
y	24	21	18	15	12	9	-171

7

a $y = -3x + 2$ **b** $y = 3x + 4$ **c** $y = 5x + 4$ **d** $y = -5x + 3$

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Applications

8

a $135\ \text{mg}$

b $t = 70$

9

a $1.4\ \text{metres/minute}$

b $y = 1.4x$

c The change in depth per minute.

d $8.4\ \text{metres}$

e $x = 9$ minutes

10

a 85

b 49 beats per minute

c 3

d $y = 3x + 49$

e The change of Mae's heartbeat in one minute.

f 109 beats per minute

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